Expected Utility And Risk Aversion

Econ 2100 Fall 2018

Lecture 12, October 8

Outline

1. Risk Aversion
2. Certainty Equivalent
3. Risk Premium
4. Relative Risk Aversion
5. Stochastic Dominance

Notation From Last Class

- A cumulative distribution function (cdf) is a function $F : \mathbf{R} \to [0, 1]$ which is nondecreasing, right continuous, and goes from 0 to 1.
- μ_F denotes the mean (expected value) of F, i.e. $\mu_F = \int x \, dF(x)$.
- δ_x is the degenerate distribution function at x; i.e. δ_x yields x with certainty:

$$\delta_x(z) = \begin{cases} 0 & \text{if } z < x \\ 1 & \text{if } z \geq x \end{cases}.$$

- Given some F, $\delta_{\mu_F} = \begin{cases} 0 & \text{if } z < \mu_F \\ 1 & \text{if } z \geq \mu_F \end{cases}$ is a probability distribution that yields the expected value of F for sure.
- Preferences are over cumulative distributions.

Risk Aversion

Definitions

The preference relation $\succsim$ is

- risk averse if, for all cumulative distribution functions F,
$$\delta_{\mu_F} \succsim F.$$
- risk loving if, for all cumulative distribution functions F,
$$F \succsim \delta_{\mu_F}.$$
- risk neutral if it is both risk averse and risk loving ($\delta_{\mu_F} \sim F$).

- DM is risk averse if she always prefers the expected value μ_F for sure to the uncertain distribution F.
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Risk Aversion: An example

Exercise

Let $\succsim$ be a preference relation on the space of all cumulative distribution functions represented by the following utility function:

$$U(F) = \begin{cases} x & \text{if } F = \delta_x \text{ for some } x \in \mathbf{R} \\ 0 & \text{otherwise} \end{cases}$$

- True of false: $\succsim$ is risk averse.

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Certainty Equivalent

Definition

Given a strictly increasing and continuous vNM index v over wealth, the certainty equivalent (CE) of F, denoted $c(F, v)$, is defined by

$$v(c(F, v)) = \int v(\cdot)\, dF.$$

- By definition, the certainty equivalent of F is the amount of wealth $c(\cdot)$ such that $c(\cdot) \sim F$.
- One can compare two lotteries by comparing their certainty equivalents.
- Unlike risk aversion, the certainty equivalent definition assumes a given preference representation (needs some utility function that represents preferences).
- The value of the certainty equivalent is related to risk aversion.

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Risk Premium

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Given a strictly increasing and continuous vNM index v over wealth, the risk premium of F, denoted $r(F, v)$ is defined by

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Risk Aversion, Certainty Equivalent, and Risk Premium

- If preferences satisfy the vNM axioms, risk aversion is completely characterized by concavity of the utility index and a non-negative risk-premium.

Proposition

Suppose $\succsim$ has an expected utility representation and v is the corresponding von Neumann and Morgestern utility index over money.The following are equivalent:

1. $\succsim$ *is risk averse;*
2. v *is concave;*
3. $r(F, v) \geq 0$;

- The proof uses Jensen's inequality.

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A function g is concave if and only if

$$\int g(x)\, dF \leq g\left(\int x dF\right)$$

- This says

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Consequences of Jensen's inequality

- Hence, $v(\cdot)$ is concave if and only if

$$\underbrace{\int v dF}_{\text{expected utility of } F} \leq \underbrace{v\left(\int dF\right)}_{\text{utility of the expected value of } F}$$

- Since v is non decreasing,

$$c(F, v) \leq \int v dF \quad \text{is equivalent to} \quad v\left(c(F, v)\right) \leq v\left(\int v dF\right)$$

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Risk Aversion, CE, and Risk Premium

$$\underbrace{\succsim \text{ is risk averse}}_{(1)} \Leftrightarrow \underbrace{v \text{ is concave}}_{(2)} \Leftrightarrow \underbrace{r(F, v) \geq 0}_{(3)}$$

We prove (1) $\Rightarrow$ (2) $\Rightarrow$ (3) $\Rightarrow$ (1). Start with (1) $\Rightarrow$ (2).

Proof.

$\succsim$ is risk averse, hence $\delta_{\mu_F} \succsim F$ for all $F \in \Delta\mathbf{R}$.

- For any $x, y \in \mathbf{R}$ and $\alpha \in [0, 1]$, let the discrete random variable Z be such that $P(Z = x) = \alpha$ and $P(Z = y) = 1 - \alpha$. Let F_Z be the associated cumulative distribution.
- By risk aversion we have:

$$v(\mu_{F_Z}) \geq \int_{\mathbf{R}} v(z) dF_Z(z)$$

$$\Leftrightarrow v(\alpha x + (1-\alpha)y) \geq \sum_{z \in \{x,y\}} v(z) P(Z = z) = \alpha v(x) + (1-\alpha) v(y)$$

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$$\begin{aligned} v(\mu_{F^{\alpha}_{x,y}}) &\geq \int v(z) dF^{\alpha}_{x,y}(z) \\ &\Rightarrow \\ v(\alpha x + (1-\alpha)y) &\geq \sum_z v(z) P(X = z) = \alpha v(x) + (1-\alpha) v(y) \end{aligned}$$

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Now prove that (2) $\Rightarrow$ (3)

Proof.

Let v be concave, and X be a random variable with cdf F.

- By Jensen's inequality:

$$v(E(X)) \geq E(v(X))$$

- Or

$$v(\mu) \geq \int v(x)dF(x) = v(CE(F, v))$$

- Since v is an increasing function, we have

$$\mu \geq CE(F, v)$$

- Thus

$$\mu - CE(F, v) \geq 0 \quad \text{i.e.,} \quad r(F, v) \geq 0$$

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$$\underbrace{\succsim \text{ is risk averse}}_{(1)} \Leftrightarrow \underbrace{v \text{ is concave}}_{(2)} \Leftrightarrow \underbrace{r(F, v) \geq 0}_{(3)}$$

Now prove that (2) $\Rightarrow$ (3)

Proof.

Let v be concave, and X be a random variable with cdf F.

- By Jensen's inequality:

$$v(\mathbf{E}(X)) \geq \mathbf{E}(v(X))$$

or

$$v(\mu_F) \geq \int v(x)dF(x) = v(c(F, v))$$

- Since v is an increasing function, we have

$$\mu_F \geq c(F, v)$$

- Thus

$$\mu_F - c(F, v) = r(F, v) \geq 0$$

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Let $r(F, v) \geq 0$ for all cdfs F.

- Then we have

$$\int_{\mathbb{R}} y \, dF_Y \geq c(F, v)$$

- which in turn implies that

$$v(\mu_F) \geq v(c(F, v)) = \int_{\mathbb{R}} v(y) \, dF(y)$$

- Hence $\delta_{\mu_F} \succsim F$ for all F, therefore $\succsim$ is risk averse.

We have shown that $(1) \Rightarrow (2) \Rightarrow (3) \Rightarrow (1)$, thus the proof is complete.

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Relative Risk Aversion

- When can we say that one decision maker is more risk averse than another?
- Relative risk aversion answers this question in a preference-based way.

Definition

Given two preference relations, $\succsim_1$ is more risk averse than $\succsim_2$ if and only if

$$F \succsim_1 \delta_x \quad \Rightarrow \quad F \succsim_2 \delta_x$$

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- If DM1 prefers the lottery F to the sure payout x, then anyone who is less risk averse than DM1 also prefers the lottery F to δ_x.
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Relative Risk Aversion

- Relative risk aversion is equivalent to: "more concavity" of the utility index, a smaller certainty equivalent, and a larger risk premium.

Proposition

Suppose $\succsim_1$ and $\succsim_2$ are preference relations represented by the vNM indices v_1 and v_2. The following are equivalent:

1. *$\succsim_1$ is more risk averse than $\succsim_2$.*
2. *$v_1 = \varphi \circ v_2$ for some strictly increasing concave function φ.*
3. *$c(F, v_1) \leq c(F, v_2)$, for all F.*
4. *$r(F, v_1) \geq r(F, v_2)$, for all F.*

Proof.

Question 5 in Problem Set 6 □

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Another Application: Asset Demand

- An asset is a divisible claim to a financial return in the future.

Asset Demand

- An agent has initial wealth w; she can invest it in a safe asset that returns \$1 per dollar invested, or in a risky asset that returns \$$z$ per dollar invested.
- The risky return has cdf $F(z)$, and assume $\int z\, dF > 1$.
- Let α and β be the amounts invested in the risky and safe asset respectively. Then, one can think of (α, β) as a portfolio allocation that pays $\alpha z + \beta$.
- The agent solves $\max \int v(\alpha z + \beta)\, dF$ s. t. $\alpha, \beta \geq 0$ and $\alpha + \beta = w$
- The first oder conditions for this optimal portfolio problem is

$$\int (z-1)v'(\alpha(z-1)+w)\, dF = 0$$

- If the decision maker is risk averse, this expression is decreasing (in α) because of the concavity of v.
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- Since concavity of v reflects risk aversion, v'' is a natural candidate measure of risk aversion.
- Unfortunately, v'' is not appropriate since it is not robust to strictly increasing linear transformations.

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Suppose $\succsim$ is a preference relation represented by the twice differentiable vNM index $v : \mathbf{R} \to \mathbf{R}$. The Arrow–Pratt measure of absolute risk aversion $\lambda : \mathbf{R} \to \mathbf{R}$ is defined by

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Suppose $\succsim$ is a preference relation represented by the twice differentiable vNM index $v : \mathbf{R} \to \mathbf{R}$. The Arrow–Pratt measure of absolute risk aversion $\lambda : \mathbf{R} \to \mathbf{R}$ is defined by

$$\lambda(x) = -\frac{v''(x)}{v'(x)}.$$

- The second derivative is normalized to measure risk aversion properly.
- Notice that by integrating $\lambda(x)$ twice one could recover the utility function.
  - How about the constants of integration?

Absolute Risk Aversion

Proposition

Suppose $\succsim_1$ and $\succsim_2$ are preference relations represented by the twice differentiable vNM indices v_1 and v_2. Then

$$\succsim_1 \text{ is more risk averse than } \succsim_2 \quad \Leftrightarrow \quad \lambda_1(x) \geq \lambda_2(x) \text{ for all } x \in \mathbf{R}$$

- This confirms that the Arrow-Pratt coefficient is the correct measure of increasing absolute risk aversion.
- One can add this to the characterizations of 'more risk averse than' that you have to prove in the homework...

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$\succsim_1$ *is more risk averse than* $\succsim_2 \quad \Leftrightarrow \quad \lambda_1(x) \geq \lambda_2(x)$ *for all* $x \in \mathbf{R}$

Proof.

We know $v_1 = \phi(v_2)$ for some strictly increasing ϕ (by the homework).

- Differentiating

$$v_1'(x) = \phi'(v_2(x))v_2'(x)$$

- Dividing v_1' by $v_1' = \phi' v_2'$ we have

$$\frac{v_1'(x)}{v_1'(x)} = \frac{\phi'(v_2(x))v_2'(x)}{v_1'(x)}$$

- Substitute the first equation

$$\frac{v_1'(x)}{v_1'(x)} = \frac{\phi'(v_2(x))v_2'(x)}{\phi'(v_2(x))v_2'(x)}$$

- or

$$\frac{v_1'(x)}{v_1'(x)} = \frac{v_2'(x)}{v_2'(x)}$$

- use the definition of Arrow-Pratt

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  $v_1'(x) = \phi'(v_2(x))v_2'(x)$ and $v_1''(x) = \phi'(v_2(x))v_2''(x) + \phi''(v_2(x))v_2'(x)$

- Dividing v_1' by $v_2' > 0$ we have

  $$\frac{v_1''(x)}{v_1'(x)} = \frac{\phi'(v_2(x))v_2''(x) + \phi''(v_2(x))v_2'(x)}{\phi'(v_2(x))v_2'(x)}$$

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  $$\lambda_1(x) = \lambda_2(x) + \frac{\phi''(v_2(x))}{\phi'(v_2(x))}$$

- Since ϕ is increasing, $\phi' > 0$, so $\lambda_1(x) \geq \lambda_2(x)$ if and only if $\phi'' \leq 0$.

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- so $\lambda_1(x) \geq \lambda_2(x)$ if and only if $\phi'' \leq 0$, i.e. ϕ is concave.

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- Because the denominator is positive, $\lambda_1(x) \geq \lambda_2(x)$ if and only if $\phi''(v_2(x)) \leq 0$ for all x.

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$$\lambda_1(x) = \lambda_2(x) + \textit{something}$$

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First Order Stochastic Dominance (FOSD)

- What kind of relationship must exist between lotteries F and G to ensure that anyone, regardless of her attitude to risk, will prefer F to G so long as she likes more wealth than less?
- In general, if a consumer's utility of wealth is increasing, but its functional form unknown, we do not have enough information to know her rankings among all distributions...but we know how she ranks some pairs; namely, those comparable with respect to the following transitive, but incomplete, binary relation.

Definition

F first-order stochastically dominates G, denoted $F \succeq_{FOSD} G$, if

$$\int v \, dF \geq \int v \, dG,$$

for every nondecreasing function $v : \mathbf{R} \to \mathbf{R}$.

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- Thus

$$\int_a^b v\, dF - \int_a^b v\, dG = \int_a^b v'(x) [G(x) - F(x)]\, dx$$

and you can fill in the blanks for a formal proof.

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Second Order Stochastic Dominance (SOSD)

If one also knows that the decision maker is risk averse (her utility index for wealth is concave), we know how she ranks more pairs.

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F second-order stochastically dominates G, denoted $F \succeq_{\text{SOSD}} G$, if

$$\int v \, dF \geq \int v \, dG,$$

for every nondecreasing concave function $v : \mathbf{R} \to \mathbf{R}$.

- If $F \succeq_{\text{SOSD}} G$, then anyone who prefers more money to less and is risk averse prefers F to G.
- By construction, the set of distributions ranked by FOSD is a subset of those ranked by SOSD

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Preferences and Lotteries Over Money

- So far, we have looked at expected utility preferences over sums of money. Dollar bills cannot be eaten, so where do these preferences come from?
- There are N commodities and $\succsim$ ranks lotteries on $X = \mathbf{R}^N_+$.
- The expected utility axioms hold: the consumer is an expected utility maximzer, and U is her expected utility function.
- One can also think of $U(\cdot)$ as a utility function in the sense of consumer theory. Let the corresponding indirect utility function be $v(p, w)$.
- Suppose all uncertainty is resolved so that the consumer learns how much money she has before she goes to the markets to buy x.
- Fix prices $p \in \mathbf{R}^N_{++}$; let $w \in [0, \infty)$ be income, and $x^*(p, w)$ the Walrasian demand. Clearly, $v(p, w) = U(x)$ for $x \in x^*(p, w)$.

How does the consumer rank lotteries over income?

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How does the consumer rank lotteries over income?

Preferences and Lotteries Over Money

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How does the consumer rank lotteries over income?

- A lottery over income π gives it a probability distribution $\pi(w)$.
- The expected utility of π is $\sum_{w \in [0,\infty)} \pi(w) v(p, w)$.
- Preferences over lotteries $\succsim$ induce preferences $\succsim^*$ over $[0, \infty)$ by $w \succsim^* w'$ if $v(p, w) \geq v(p, w')$.
- The indirect utility function $v(p, w)$ is an ordinal concept.
- How do preferences $\succsim^*$ change with prices p?

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How does the consumer rank lotteries over income?

- A lottery over income is $(p_1, w_1; \ldots; p_K, w_K)$ that pays w_k with probability p_k.
- The expected utility of this lottery is $\sum_{k=1}^{K} p_k v(p, w_k)$.
- The indirect utility function $v(p, w)$ is cardinal in w.
- Therefore, preferences under uncertainty over income are cardinal.

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How does the consumer rank lotteries over income?

- A lottery over income is (p, L) where L is a lottery over $[0, \infty)$.
- The consumer's utility from (p, L) is $\int v(p, w) \, dL(w)$.
- The indirect utility function $v(\cdot)$ is von Neumann-Morgenstern.
- Does this mean that $(p, L) \succsim (p, M)$ if and only if $\int v(p, w) \, dL(w) \geq \int v(p, w) \, dM(w)$?

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How does the consumer rank lotteries over income?

- A lottery over income is given by a cdf F that has support in $[0, \infty)$.
- The consumer ranks lotteries according to the expected value of $v(p, w)$ with respect to F.
- The indirect utility function $v(p, w)$ is von Neumann-Morgenstern utility function over income lotteries.

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How does the consumer rank lotteries over income?

- A lottery over income $\pi(w)$ is a probability distributions on $[0, \infty)$.
- The expected utility of π is $\sum_{y \in support(\pi)} \pi(w)\, v(p, w)$; and

$$\pi \succsim \rho \Leftrightarrow \sum_{y \in support(\pi)} \pi(w)\, v(p, w) \geq \sum_{y \in support(\pi)} \rho(w)\, v(p, w)$$

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